Problem Set 1

AI 501: Mathematics for Artificial Intelligence

Fall 2026

Problem 1. A *Boolean n-vector* has every entry equal to 0 or 1, and is used to record whether each of n conditions holds (1 meaning the condition holds). A second, equally common encoding of the same information uses the values -1 and $+1$ instead of 0 and 1: for example, the Boolean vector $(0, 1, 1, 0)$ would be written in this alternate encoding as $(-1, +1, +1, -1)$. If x is a vector in the $\{0, 1\}$ encoding and y encodes the same information in the $\{-1, +1\}$ encoding, write y in terms of x using vector notation, and also write x in terms of y.

Problem 2. A 20-vector s records whether each of 20 possible symptoms is present in a patient, with $s_i = 1$ if symptom i is present and $s_i = 0$ otherwise. Using vector notation, express:

(a) the total number of symptoms the patient has;

(b) the statement "the patient has exactly five of the first ten symptoms."

Problem 3. Suppose the n-vector w is the word-count vector for some document, built from a dictionary of n words (assume every word in the document is one of the dictionary words).

(a) What is $\mathbf{1}^T w$?

(b) What does the entry $w_{282} = 0$ tell you about the document?

(c) Let h be the n-vector giving the *histogram* of word frequencies, i.e., h_i is the fraction of the words in the document equal to dictionary word i. Express h in terms of w, assuming the document has at least one word.

Problem 4. A company holds cash in five currencies: USD, RMB, EUR, GBP, and JPY. The amounts are given by the 5-vector c (so c_2 is the number of RMB held; negative entries represent liabilities). Using vector notation, express the total value of this holding in USD. Be sure to introduce and define any additional vector(s) your expression needs.

Problem 5. Let $w = (0, 2, -1)$ and $z = (-3, 2)$. For each expression, evaluate whether the operation is valid, and if so, give its value.

(a) $w + z$

(b) $w^T z$

(c) $z^T w$

(d) wz

Problem 6. Suppose each of the vectors b_1, b_2 is a linear combination of a_1, a_2, and c is a linear combination of b_1, b_2. Show that c must then be a linear combination of a_1, a_2.

Problem 7. An algorithm predicts a stock's price tomorrow $\hat{z}_{t+1}$ from the last $M = 10$ values using $\hat{z}_{t+1} = (z_t, z_{t-1}, \ldots, z_{t-9})^T \beta$. For each choice of the coefficient vector β below, describe in plain English (no vectors, inner products, etc.) what the algorithm is doing. You may use words like "yesterday" or "today."

(a) $\beta \approx e_1$

(b) $\beta \approx 2e_1 - e_2$

(c) $\beta \approx e_6$

(d) $\beta \approx 0.5e_1 + 0.5e_2$

Problem 8. A student's record is a 10-vector r: $r_1, \ldots, r_8$ are homework grades (each on a 0–10 scale), r_9 is the midterm grade (0–120 scale), and r_{10} is the final exam grade (0–160 scale). The course score s, on a 0–100 scale, weights homework 25%, the midterm 35%, and the final 40%. Find the 10-vector w (as a linear combination / weighted sum of the entries of r) so that $s = w^T r$. Give entries of w to four decimal places.

Problem 9. Let x_1, x_2 be two points in $\mathbb{R}^2$, and consider combinations of the form $y = \alpha_1 x_1 + \alpha_2 x_2$. Describe, and sketch if you like, the set of all such points y under each of the following restrictions on the coefficients:

(a) no restriction on α_1, α_2 (a general linear combination);

(b) $\alpha_1, \alpha_2 \geq 0$ (a *conic* combination);

(c) $\alpha_1 + \alpha_2 = 1$ (an *affine* combination);

(d) $\alpha_1, \alpha_2 \geq 0$ and $\alpha_1 + \alpha_2 = 1$ (a *convex* combination).

Problem 10. Let x_1, x_2, x_3 be three points, and consider their affine combination $y = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3$ with $\alpha_1 + \alpha_2 + \alpha_3 = 1$. Eliminate α_3 using the constraint and describe, geometrically, the set of all points y obtainable this way when x_1, x_2, x_3 are not collinear.

Problem 11. Let v, w be nonzero vectors in $\mathbb{R}^3$. Describe geometrically the set of all linear combinations $cv + dw$ (this set is called the *span* of $\{v, w\}$) in the case where v and w point along the same line through the origin, and in the case where they do not.

Problem 12. Let

$$u = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}, \quad w = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Find scalars c, d, e such that $cu + dv + ew = b$, i.e., show that b is in the span of $\{u, v, w\}$.

Problem 13. In $\mathbb{R}^5$, is the vector $(8, 5, -2, 3, 1)$ a linear combination of

$$v_1 = (1, 0, 0, 0, 0), \quad v_2 = (1, 1, 0, 0, 0), \quad v_3 = (1, 1, 1, 0, 0), \quad v_4 = (1, 1, 1, 1, 0)?$$

That is, is $(8, 5, -2, 3, 1)$ in the span of $\{v_1, v_2, v_3, v_4\}$? Answer without doing any elimination.

Problem 14. Let α, β be scalars and let a, b be orthogonal n-vectors. Express $\|\alpha a + \beta b\|$ in terms of $\|a\|, \|b\|$, α, β.

Problem 15. Suppose x is an n-vector with $n > 1$. Which of the following best describes x^2 (the vector x raised to the second power, with no transpose)?

(a) it is the inner product of x with itself;

(b) it is the vector with entries x_i^2;

(c) it makes no sense.

Choose one and justify your choice.

Problem 16. Let x be any vector. For each subproblem, choose the correct option.

(a) $\|x\| = -\|-x\|$ holds: (i) always (ii) sometimes (iii) never.

(b) $\|(x, x)\|$ (the norm of x stacked on top of itself) equals: (i) $\|x\|$ (ii) $2\|x\|$ (iii) $\sqrt{2}\,\|x\|$ (iv) $\frac{1}{2}\|x\|$.